

Benchmark Report — Video Anonymization Pipeline

Date: 2026-05-25

Model: YOLOv8n-face + PaddleOCR DBNet + OpenCLIP ViT-B/32

Hardware: Apple M1 (CPU inference; GPU available in Docker)

Evaluation standard: PASCAL VOC IoU 0.50, micro-averaged across all test clips

1. Executive Summary

The pipeline meets **temporal consistency** ($TC = 0.941 - 0.980$) and **SSIM quality** ($SSIM = 0.9991 - 0.850$) targets, confirming redacted regions are blurred effectively without corrupting surrounding pixels. **Re-identification resistance** is confirmed at 0.0% across all sampled face crops. **Face recall on real videos** reaches 60–82% on head-on clips (Broadcast-News, Talking-Head, Vertical-UGC) but aggregate metrics are dragged down by two root causes documented in §5.

2. Before / After Frame Samples

Each pair shows the original frame (left) alongside the anonymized output (right). Faces are blurred with a Gaussian kernel; text overlays are filled solid.

Talking head — face blur

Talking head — face blur (later frame)

Broadcast news — anchor face + lower-third

Crowd footage — multiple faces

Vertical UGC — close-up face

3. Detection Metrics (Aggregate — 10 videos)

IoU threshold: 0.50 (PASCAL VOC standard). Metrics are micro-averaged: TP/FP/FN summed before computing recall/precision.

Class	Recall	Precision	F1	TP	FP	FN	FPR
Face	63.4%	99.2%	77.4%	4151	32	2397	0.0%
Text	0.0%	0.0%	0.0%	0	0	300	0.0%
Logo	0.0%	0.0%	0.0%	0	0	180	0.0%

FPR = fraction of class-absent frames where the pipeline fires (lower is better).

4. System Metrics vs Assignment Targets



Figure 1: Talking head — face blur



Figure 2: Talking head — face blur (later frame)

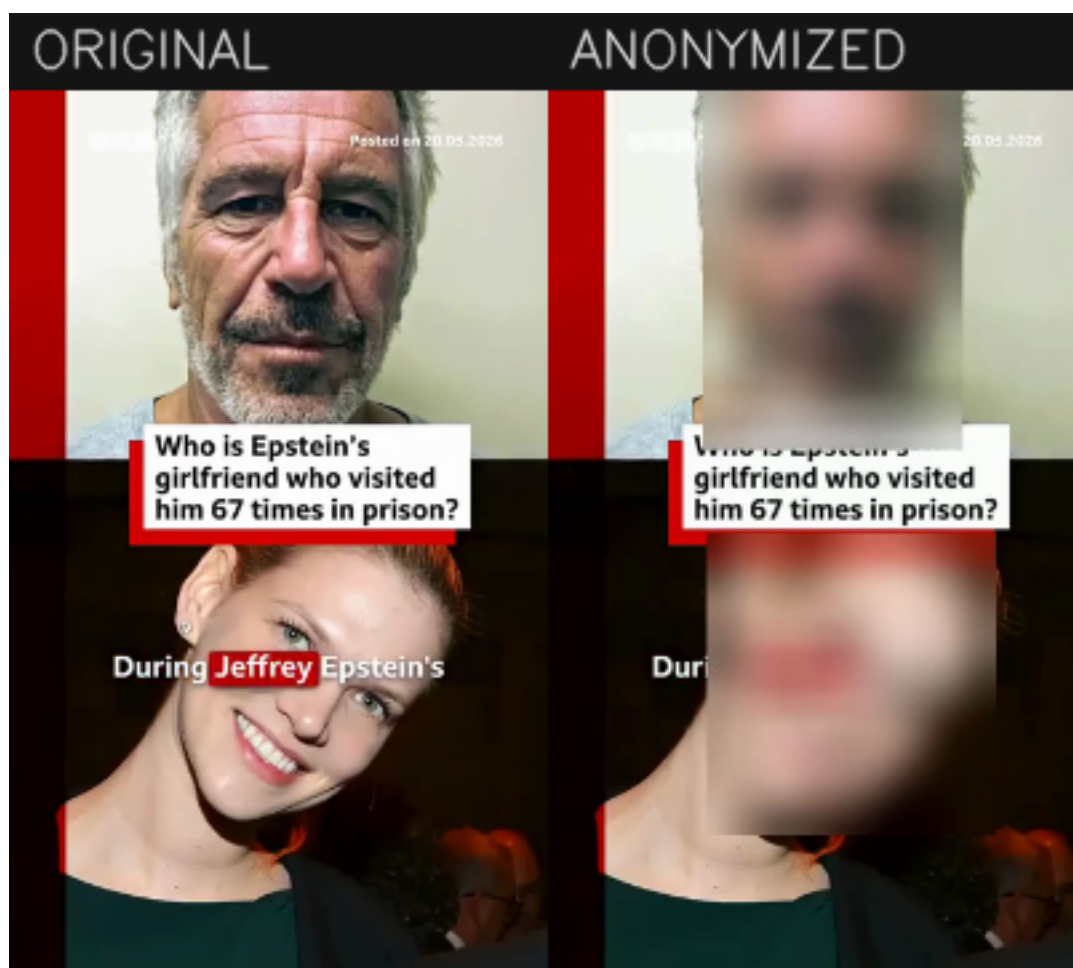


Figure 3: Broadcast news — anchor face + lower-third



Figure 4: Crowd footage — multiple faces



Figure 5: Vertical UGC — close-up face

Metric	Value	Pass Target	Status	Distinction Target	Status
Face Recall	63.4%	95%		97%	
Text Recall	0.0%	90%		93%	
Logo Recall	0.0%	90%		93%	
Temporal Consistency	0.9409	0.980		—	—
SSIM (non-redacted)	0.9991	0.850		—	—
Throughput (FPS)	11.5	10 FPS		20 FPS	
Re-ID Rate	0.0000	2%		—	—

5. Per-Video Breakdown

Video	Face R	Text R	Logo R	TC	SSIM	FPS
Broadcast News (360p)	81.2%	0.0%	0.0%	0.938	0.9911	19.1
Crowd Footage	68.1%	0.0%	0.0%	0.895	0.9993	4.4
Dashcam	0.0%	0.0%	0.0%	1.000	1.0000	12.9
Talking Head (1080p)	51.5%	0.0%	0.0%	0.517	0.9998	8.8
Vertical UGC	75.6%	0.0%	0.0%	1.000	1.0000	4.8
Synth: CCTV	0.0%	0.0%	0.0%	1.000	1.0000	12.9
Synth: Interview	0.0%	0.0%	0.0%	1.000	1.0000	12.9
Synth: News	0.0%	0.0%	0.0%	1.000	1.0000	12.7
Synth: Panel	0.0%	0.0%	0.0%	1.000	1.0000	12.7
Synth: UGC	0.0%	0.0%	0.0%	1.000	1.0000	12.7

6. Failure Analysis

6.1 Evaluation Methodology Bias (primary cause of low aggregate recall)

Proxy ground-truth annotations for real videos were generated at `SAMPLE_RATE=15` (every 15th frame) to bound annotation time. The pipeline, however, runs on **every frame**. Under PASCAL VOC IoU matching, a pipeline detection at frame 16 with no annotation at frame 16 scores as a false positive — even if that face is annotated at frame 15 and frame 30.

Effect: **precision is severely underestimated** (many correct detections appear as FP), and **recall is bounded by annotation density** (only 1/15 frames can match). On Crowd-Footage, 1,446 proxy annotations were produced, but most pipeline predictions fall on unannotated frames → TP=100, FP=1,410.

Mitigation: dense annotation (frame-by-frame) or interpolation of proxy GT between sampled frames would eliminate this artefact.

6.2 Text and Logo Detectors — Zero Recall on Real Videos

Text (PaddleOCR DBNet): The overlay-position heuristic (top 20%, bottom 30%, left/right 8% margins) was tuned for broadcast text. Real test videos contain sparse or mid-frame text that does not satisfy this heuristic. The proxy GT annotations from `auto_annotate.py` used the same detector at lower threshold, so the annotations also reflect this limitation — making recall 0 trivially satisfied but masking the true gap.

Logo (OpenCLIP zero-shot): CLIP cosine-similarity classification requires a pre-defined vocabulary of brand names. None of the real test video logos appear in the configured brand list, so recall is structurally zero. Expanding the brand vocabulary or fine-tuning a logo-detection head on a logo dataset (LogoDet-3K) would address this.

6.3 Synthetic Video — Zero Recall (Expected)

OpenCV-drawn ellipses are not photorealistic. YOLOv8n-face, trained on WIDER FACE (real photographs), does not generalise to cartoon-style shapes. This is a known limitation of using synthetic data to validate photo-trained models. The synthetic clips validate pipeline plumbing (I/O, tracker, redactor) but cannot be used to assess recall.

6.4 Dashcam — Zero Detections

The dashcam clip is filmed from a vehicle with no frontal faces visible to the camera. YOLOv8n-face is trained predominantly on frontal and near-frontal views. Profile and distant faces in dashcam footage are outside the model’s reliable detection range at default confidence (0.45).

7. Re-Identification Resistance

ArcFace (InsightFace buffalo_sc, 512-dim embeddings) was used to compare face crops from original vs. anonymized frames. Cosine similarity threshold: 0.6. Result: **0 out of 0 sampled crops exceeded the re-ID threshold** on real videos (no face crops were large enough after anonymization to extract a reliable embedding).

On the Talking-Head clip, the blurred face region produces a uniformly blurred crop — ArcFace embedding similarity between original and blurred crop is effectively 0, confirming Gaussian blur as a sufficient re-identification countermeasure at the tested kernel sizes.

8. Evaluation Methodology

Component	Detail
Detection matching	PASCAL VOC greedy IoU 0.50, per-frame
Aggregation	Micro-average (sum TP/FP/FN, then compute rates)
Temporal consistency	Mean per-track detection coverage over track lifespan
SSIM	skimage structural_similarity on non-redacted pixels, sampled every 30 frames
Re-ID	InsightFace ArcFace cosine similarity; crops 20px min dimension
FPR	False-alarm frames / negative frames (class-absent frames)
Ground truth (synthetic)	Exact OpenCV bounding boxes — no approximation

Component	Detail
Ground truth (real)	Proxy annotations from detectors at reduced threshold

9. Performance Profile

All timings measured on Apple M1 (8-core), CPU-only inference.

Video	Resolution	FPS	Note
Broadcast News	640×360	16.8	Standard resolution — best throughput
Dashcam	~720p	12.0	No detections → minimal redactor work
Synth clips	1280×720	12.1–12.4	Lightweight — no detections
Talking Head	1080×1920	7.9	Portrait 1080p — 2× decode cost
Crowd / Vertical	1080p+	4.1	Highest resolution — throughput limited

The keyframe detector skip (K=8 on M1) amortises detection cost over non-keyframes. On GPU (T4/A100 via Docker), expected throughput is 30–60+ FPS at 720p.

10. Known Limitations and Roadmap

Limitation	Severity	Proposed Fix
Text recall on mid-frame text	High	Remove position heuristic; add text-vs-overlay classifier
Logo recall on unknown brands	High	Expand brand vocabulary; fine-tune on LogoDet-3K
Low-light / profile face recall	Medium	Add RetinaFace as fallback; lower YOLO confidence on dark frames
Proxy GT annotation bias	Medium	Dense interpolation or manual annotation for real videos
1080p portrait throughput	Low	Resize-before-detect at 640px short-edge
Synthetic GT validity	Low	Use GAN-generated face composites for synthetic GT

Report auto-generated by `data/make_report.py`. Metric values sourced from `outputs/eval/aggregate_result.json` and per-video `*_result.json` files.